Privacy Enhanced Authentication, ZKP & Proof of Personhood

IKP session 2-6 · authentication that proves the claim without disclosing the person

Blockchain Governance Initiative Network

Mitchell Travers (chair) · World as main discussant · contributors to the IKP proof-of-personhood paper

BGIN Block #15 | Georgetown University Capitol Campus, Washington, D.C. | 16 October 2026, Room B, 11:30–13:00, then the World lunch session 13:00–14:00 | hybrid

Builds on the Block 14 reports *Privacy Enhanced Authentication and Key Management* and *Proof of Personhood* and the open IKP meetings since · disclosure: the chair also co-chairs the ToIP DTG ZKP Task Force; observations on that work are personal and from public materials

This Session — What It Is For

Privacy-enhanced authentication at a high level, the proof-of-personhood taxonomy and the distinctions it has to keep, verifiable trust agents acting for a person, and where zero-knowledge proofs do the work. Twenty minutes for contributors, then open discussion into the IKP proof-of-personhood paper.

1. Why now: agentic AI raises the cost of not knowing a counterparty is human
2. Prove the claim, not the identity: minimal disclosure, unlinkability, what each proof proves, where the biometric lives, what revocation means when the binding cannot be rotated
3. The taxonomy, with World as main discussant: unique human versus specific human, humanity versus uniqueness, linkability scope, strength expression, continuity, agent authority
4. Verifiable trust agents: an agent acting for a person carrying personhood assurance with delegated scope
5. ZKP where it earns its place: predicates that carry strength, client-side proving on everyday phones, post-quantum readiness of the stack
6. Contributions (20 min); then the gaps nobody owns yet; then next steps into the paper, public comment, and the ISO TC 307 Technical Report path

🔗 **Takeaway:** Leave with room comments on the paper's structure, a contributor list, and the unclaimed gaps confirmed as IKP's contribution.

Why Now — Agentic AI Raises the Cost of Not Knowing

When an AI system rather than a person can hold a wallet, pay, post and transact, every counterparty has to ask a question that used to be free: **is there a human here, and on whose authority is this agent acting?** The answer must not cost the human their privacy.

Block 14 left two reports

- *Privacy Enhanced Authentication and Key Management*: one system holds the keys, another composes the transactions; AI as a separate process from the key holder, with scoped, revocable access
- *Proof of Personhood*: the three-graph model, the dual-agent separation, the human stack, and the ISO TC 307 Technical Report path — Block 15 named for public comment

Since then

- IKP research pathway on ZK proofs for biometric authentication (July); open, topic-focused IKP meetings on the taxonomy
- A working note mapping three converging efforts: World's Proof of Human framework, the ToIP DTG credential and proof-of-liveness work, this paper
- Client-side ZK proving on everyday phones became real (ProveKit)

🔗 **Takeaway:** The question is no longer whether personhood can be proven privately, but what a given proof proves, and to whom.

Prove the Claim, Not the Identity

What each proof proves

Proof of *a* human (uniqueness) is not proof of *a specific* human (identity). A relying party should be told which one it is getting, at what strength, against which adversary — and what the proof explicitly does not establish.

Where the biometric lives

On the device, signed by the capture hardware and held by the person; uniqueness checked without an identity lookup (sMPC in World's design); government ID matched locally if at all. The architecture decides the harm if it fails.

Revocation of what cannot rotate

A key can be rotated. An iris cannot. Revocation for a credential bound to a biometric means revoking the *binding* and the *enrolment*, not the person — and saying so in the taxonomy.

Minimal disclosure and unlinkability are the two privacy properties every pattern here is scored on. Verifiability is not truth: a proof can be perfectly verifiable and still establish less than the relying party assumes.

🔗 **Takeaway:** A taxonomy that cannot say which of these a scheme delivers cannot be used to certify one.

Three Efforts, One Stack — the Convergence Map

Three efforts converged on the same structure independently. The friction is almost entirely at the seams between layers, which is where a working group earns its keep.

Layer	Question it answers	Who is working here
Assurance semantics	What class of claim, at what strength, on what evidence; how it maps to NIST xAL and ISO 29115	World *Proof of Human* framework (v0.13, Aug 2026: ten categories, four dimensions)
Predicate semantics	What a proof establishes, what it does *not*, what it discloses, to whom, against which adversary, for how long	ToIP DTG ZKP Task Force, proof-of-liveness requirements v0.4
Mechanism	Which credential, protocol and proof construction carries the claim	DTG Core Credentials WD01; First Person Network
Formal security	What the claim provably guarantees against a defined adversary	*A Cryptographic Framework for Proof of Personhood* (ePrint 2026/333)
Governance & policy	Who may assert this, in what domain, under what regulatory consequence	**BGIN IKP**; trust registries; ISO TC 307

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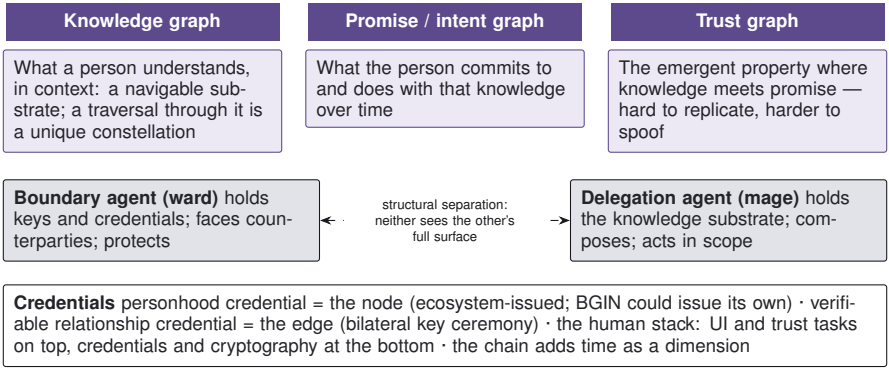
🔗 **Takeaway:** Published standalone, a new taxonomy is a fourth competing vocabulary. Published as a layer bound to the predicates beneath it, it is what makes the others interoperable.

Five Distinctions the Taxonomy Has to Keep

Distinction	Why it is load-bearing
Humanity $\neq$ uniqueness	Two different claims, routinely collapsed into one credential. Liveness plus presence plus absence-of-automation is not the same as "one person, once, in this domain".
Linkability scope	Adopt the architecture (pairwise, domain-scoped, global), then parameterise the dimension. A uniqueness assertion that survives across domains is a correlation key.
Strength expression	The headline gap: how assurance is *stated* rather than asserted — ordinal scale, per-category scale, or a profile-defined vector a relying party can actually consume.
Continuity	Continuity with enrolment is a property of the carrier and its controls, not of the primitive. Enrolment poisoning is its adversary.
Agent authority	An agent acting for a human: the field list exists in the credential work; it needs completing with scope, delegation and revocation.

🔗 **Takeaway:** World is the main discussant here because its framework is strongest on strength expression and on separating humanity from uniqueness — the two places the existing standards are

The Block 14 Model — Three Graphs, Two Agents



🔗 **Takeaway:** The knowledge substrate never touches the wallet. The person is proven at the overlap of what they know, what they promise, and the trust that emerges between two agents that cannot fully see each other.

Verifiable Trust Agents — Personhood with Delegated Scope

An agent acting for a person can carry personhood assurance with a declared scope. The question for the taxonomy: does the principal need to be *identified*, or only *assured as human* with delegated authority?

What the agent carries

- A personhood credential as the node; relationship credentials as the edges it may traverse
- Delegation scope: what it may read, what it may share, on whose authority, until when
- Revocation that reaches the delegation, not the person

The written contribution to this Block

- *Separation of Duties for Personal AI Agents* (discussion paper proposing a work item): two agents with disjoint duties and structural context erasure between them
- Separation classified by **reachability**, not by rule: policy-enforced if some permitted sequence can reconstruct the shared origin; structural if none can
- Deliverables: a documentation profile under which every separation claim names its preconditions, adversary class and date. Eight open questions, none asserted

Zero-Knowledge Proofs Where They Earn Their Place

Predicates that carry strength

A proof should disclose the assurance class and policy identifier without leaking the issuer. Open: whether issuer–verifier collusion resistance is achievable at all once schema rarity, status traffic and timing are counted.

Proving on the phones people own

ProveKit (World, Atheon, Reilabs, Nethermind): Noir circuits proven on-device, verified on chain or in a browser. iPhone SE 3: 2–3 s; 2 GB Android: under 30 s; under 500 MB peak memory. “Trusted setup is a distribution problem before it is a trust problem.”

Post-quantum readiness of the stack

Personhood credentials, relationship credentials and ZK proofs over biometrics all rest on primitives in the PQC migration’s scope. Lattice-based proof systems are arriving; the credential stack has to be migratable, not just the signatures.

Not a ZKP-only session: the proof layer is one row of the stack. It does the work only where the predicate it proves has been written down with its negative space — what it does not establish.

🔗 **Takeaway:** “PQC-ready” and “privacy-preserving” are the same kind of claim: testable only against a published definition. Thursday’s register and today’s taxonomy are two halves of that definition.

Gaps Nobody Owns — Candidate IKP Work Items

Gap	Why it is BGIN's
Coercion, duress, credential rental	A human can be present, live, unique and cryptographically flawless — and still a rented proxy. Fencing it off as "not established" is not treatment. Governance and economics more than cryptography.
Threat model per assurance category	Sybil (uniqueness) · injection and deepfake (presence) · enrolment poisoning (continuity) · credential theft (control) · duress (action) · farming. ISO SC 27 needs the adversary column; the drafting discipline exists, its application does not.
Governance of the domain	"Unique within a declared domain" needs someone authorised to declare and govern it. Who accredits a context authority? What happens when two disagree about the same human? Appeal when a human is falsely denied.
Regulatory interface	An unlinkable-by-design personhood credential and a sanctions-screening duty that presumes identifiability are on a collision course. Neither ToIP nor World will write this. Session 2-9 this afternoon is where it lands.
Global registry harms	A uniqueness assertion that survives across domains is a correlation key: exclusion, state coercion, irreversible biometric enrolment, function creep.

🔗 **Takeaway:** Unclaimed across both ToIP workstreams, unaddressed by NIST, squarely in BGIN's competence. The paper takes them as its contributed section.

Session 2-6 — Discussion Items

1. Should assurance strength be a **single ordinal scale**, per-category scales, or a profile-defined vector? What does a relying party actually consume?
2. Is “humanity” a **primitive claim**, or the union of liveness, presence and absence-of-automation? Is the third even provable? If it decomposes, strength attaches per component.
3. For **agent authority**: does the human principal need to be identified, or only assured as human with delegated scope?
4. How does any personhood claim survive **coercion and credential rental** — and if it cannot, how should the taxonomy say so?
5. Who **accredits a context authority**, what is the appeal path when a human is falsely denied uniqueness, and who supervises?

Every answer in the form the predicate work already uses: every parameter bound, adversary and horizon declared, negative space written.

Session 2-6 — Where It Goes

Room comments go into the IKP proof-of-personhood paper; the paper goes out for public comment; the standards path is an ISO TC 307 Technical Report — a survey of the state of the world, one voting round, not a four-year International Standard.

The paper

- Three-graph model and dual-agent separation (Block 14)
- Assurance decomposition as a contributed section; the convergence mapping as an annex
- The unclaimed gaps as the BGIN-specific contribution
- PQC migration of the credential stack as a requirement, not an afterthought

The afternoon

- 13:00–14:00, Room B — lunch session with World: proof of a unique human in practice, and the client-side ZK tooling behind it; questions from the room feed the taxonomy
- 14:00–14:45, Room B — Offline Key Management (Cyber + IKP): the same separation principle, applied to cold storage
- 15:40–17:10, Room A — the regulatory interface, next

🔗 **Takeaway:** Bring something half-built from real implementation to the standards body; do not start from scratch in committee. The room's job is to make the half-built thing honest.

How to Engage

- **Read** the two Block 14 meeting reports on the Privacy Enhanced Authentication hub.
- **Add research questions** to the ZKP / biometric authentication pathway thread on the IKP Discourse.
- **Join** the open, topic-focused IKP meetings on the taxonomy between Blocks.
- **Comment** on the proof-of-personhood paper when it is released for public comment; contributors are credited in it.
- **Stay** for the lunch session with World, 13:00 in this room: proof of a unique human in practice, and the client-side ZK tooling behind it.

This session
bgin-global.org/events/20261015-block15/sessions/2-6

Project hub
bgin-global.org/projects/privacy-enhanced-auth

BGIN Discourse
bgin.discourse.group

🔗 **Takeaway:** The paper goes out for public comment after this Block; the room's comments decide what it says about strength, uniqueness and agent authority.

Prove the claim, not the identity.

The person is proven at the overlap: what they know, what they promise, and the trust that emerges between two agents that can never fully see each other.

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